# Example Exam
# Medical Image Analysis

Student name: ______________________________

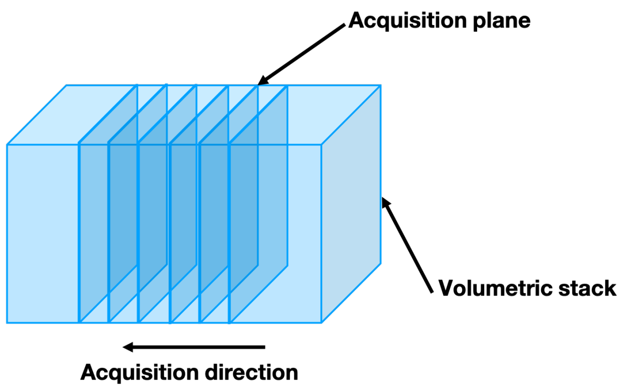
Immatriculation number: ______________________________

## Exam Questions

Note: All the questions have equal weight.

1. In 2D acquisitions, 2D image slices are acquired in one pre-defined direction one after the other, and the final 3D volume is created by stacking the 2D slices. For instance, in axial acquisitions 2D slices are acquired in the transverse plane, while in coronal acquisitions, 2D slices are acquired in the coronal plane. Figure below illustrates these planes. The voxel sizes of 2D acquisitions can be provided based on the acquisition direction, as in-plane and through-plane voxel sizes, where the first is represented with two values (pixel size in the imaging plane) and the latter corresponds to a single value (length of the voxel in the third dimension). Based on this please compute the following

   (a) An axial acquisition with voxel size in-plane $0.6 \times 0.6\ mm^2$ with 320 pixels in both directions and voxel size through-plane $3.5\ mm$ with 150 pixels is acquired. What is the **field-of-view** in the head-to-toe, left-to-right and anterior-to-posterior directions, respectively?



**Solution:** Answer: $525 \times 192 \times 192\ mm^3$

(b) A sagittal acquisition with voxel size in-plane $0.75 \times 0.75\ mm^2$ with 256 pixels in both directions and voxel size through-plane $2.5\ mm$ with 210 pixels is acquired. What is the **field-of-view** in the head-to-toe, left-to-right and anterior-to-posterior directions, respectively?

**Solution:** Answer: $192 \times 525 \times 192\ mm^3$

(c) A coronal acquisition with voxel size in-plane $1.0 \times 1.0\ mm^2$ with a field-of-view of $192\ mm$ in both directions and voxel size through-plane of $2.5\ mm$ with field-of-view of $275\ mm$. What are the **number of pixels** in the head-to-toe, left-to-right and anterior-to-posterior directions, respectively?

**Solution:** Answer: $192 \times 192 \times 110$ voxels

2. For a given image segmentation problem, with I indicating the image and S the segmentation, which of the following maximization problems corresponds to the Maximum-A-Posteriori estimation in the probabilistic formulation? Please circle the correct choice.

$$\max_S \log p(I|S) \qquad \max_S \log \left[p(I|S)p(S)\right]$$

**Solution:** Answer: Right one

3. We discussed that the following two optimization problems are closely related and equivalent under certain cases. The first problem is derived from a probabilistic model while the second one is energy-based.

$$\max_S \log p(I|S) + \log p(S) \qquad \min_S \mathcal{D}(I,S) - \lambda\mathcal{R}(S)$$

Please indicate which term corresponds to which term in the two formulations.

**Solution:** Answer: $\log p(I|S)$ <<>> $\mathcal{D}(I,S)$ and $\log p(S)$ <<>> $\lambda\mathcal{R}(S)$

4. Please indicate whether the statement is true or false by circling the correct choice in the each of the following.

(a) **TRUE** False Nyul's intensity normalization is a non-linear method.

(b) True **FALSE** Min-max normalization when used with 0 and 100 percentiles is particularly robust to intensity outliers.

(c) True **FALSE** Matching mean and standard deviations of two images can remove the intensity differences due to bias fields in Magnetic Resonance Imaging.

(d) True **FALSE** If an image has two different types of pathologies, one with hyper-intense (higher then normal tissue) and the other with hypo-intense (lower than normal tissue) intensities, then in min-max normalization, their effects would cancel out, resulting in zero net adverse effect during matching the intensity profile to an image from a healthy subject.

5. Mixture models and Expectation-Maximization (EM) segmentation is an essential building block for many of the recently proposed algorithms, as well as current software suites for analyzing brain images. For a given image with N pixels and a single intensity value per pixel, how many parameters does a Gaussian Mixture Model (GMM) with 4 components have?

**Solution:** Answer: 4 x 3 = 12.

6. The EM algorithm is an optimization algorithm with two alternating steps: Expectation and Maximization. Letting $c(x)$ denote the class assignment for each pixel at $x$, please briefly describe what the two steps of the EM algorithm do in terms of the class assignments and parameters of the mixture model. You do not need to write down the equations.

**Solution:** Answer: Expectation step computes posterior distributions given model parameters. This corresponds to soft class assignments. Maximization step optimizes for the model parameters given the current soft class assignment.

7. Which of the followings statements on similarities and differences between the K-Means segmentation algorithm and EM algorithm with Gaussian Mixture Model (GMM) are **inaccurate**?

○ (A) EM algorithm with GMM uses probabilistic (soft) class assignments while K-Means uses deterministic (hard) class assignments.

○ (B) EM algorithm with GMM estimates the mean vector and the covariance matrices of the likelihood model while K-Means algorithm only uses the mean.

○ (C) EM algorithm with GMM does not share the sensitivity of the K-Means algorithm to the number of components set by the user.

○ (D) Initialization is important for both the EM with GMM and K-Means algorithms.

**Solution:** Answer: C

8. The historical evolution of the machine learning approaches for analyzing medical images, from univariate statistical models to multi-layered convolutional neural networks, has seen a trend towards reducing the human involvement in the use and design of features. Which of the following statements on this evolution is **incorrect**?

   ◯ (A) Univariate statistical models used hand-crafted and often difficult to extract features with the goal of identifying statistical relationships between features and labels.

   ◯ (B) Multivariate predictive models, such as Support Vector Machines and Logistic Regression, reduced the effort for extracting features compared to univariate models by using multiple hand-crafted features in combination.

   ◯ (C) Models using internal feature selection, such as Random Forest, was able to reduce the feature extraction effort by being able to use lots of primitive features that are easier to extract rather than a couple of hand crafted features as was the case in univariate statistical and multivariate predictive models.

   ◯ (D) Multi-layered neural networks automatically extracts features from raw data at the expense of increasing the number of model parameters.

   **Solution:** Answer: B

9. Context is crucial for segmenting medical images. As can be seen in the example below, segmenting an image pixel becomes substantially easier if image information around the pixel is available. While the anatomy in the left most image patch is not predictable based on the patch itself, information on the right most image provides enough information to identify it. Which of the following algorithms do not use context at all while determining the class assignment of a pixel?

   ◯ (A) Expectation-Maximization with pixel-wise Gaussian Mixture Models

   ◯ (B) Atlas-based segmentation

   ◯ (C) Random Forest

   ◯ (D) Convolutional Neural Networks

   **Solution:** Answer: A

10. What is the difference between logistic regression and a single neuron with a sigmoid activation function?

   ◯ (A) The equation for logistic regression does not include a bias term.

   ◯ (B) A single neuron can model more complex relationships between the input data and the labels.

   ◯ (C) Logistic regression can model more complex relationships between the input data and the labels.

   ◯ (D) Only logistic regression allows a probabilistic interpretation.

   ◯ (E) There is no difference.

   **Solution:** Answer: E

11. The LeNet-5 (proposed in 1998) is a very simple, yet at the time ground-breaking convolutional neural network for classifying hand-written digits. It takes input images of size $32 \times 32$ with one input channel (i.e. grey scale images) and has the following architecture:

Convolutional part:

- *Conv1*: 6 kernels with kernel size $K = 5 \times 5$, stride $S = 1 \times 1$ in each direction, and no padding $P = 0$.
- *Pool1*: Max pooling with $K = 2 \times 2$ and $S = 2$
- *Conv2*: 16 kernels with $K = 5 \times 5$, stride $S = 1$ and $P = 0$.
- *Pool2*: Max pooling with $K = 2 \times 2$ and $S = 2$

Fully connected part:

- Fully connected layer with 120 hidden neurons
- Fully connected layer with 84 hidden neurons
- Fully connected layer with 10 output neurons (because there are 10 digits)

The following questions are only about the *convolutional part* of the above network. Circle True or False:

(a) **TRUE** False The global receptive field of a neuron in *Pool2* is $16 \times 16$.

(b) True **FALSE** The global stride of a neuron in *Pool2* is $2 \times 2$.

(c) **TRUE** False The total number of convolutional kernel weights (excluding the biases) in *Conv1*, *Pool1*, *Conv2* and *Pool2* is $5 \cdot 5 \cdot 1 \cdot 6 + 5 \cdot 5 \cdot 6 \cdot 16 = 2550$.

(d) True **FALSE** If grey-scale input images of size $64 \times 64$ were used instead of the $32 \times 32$ images, the number of convolutional kernel weights in part (c) would be 4 times larger.

12. What are advantages of convolutional layers over fully connected (dense) layers such as the ones used in multi-layer perceptrons (multiple answers are possible):

○ (A) A single convolutional layer has a much larger receptive field than a fully connected layer.

○ (B) In some cases, convolutional layers allow to drastically reduce the number of parameters needed in a single layer.

○ (C) In contrast to fully connected layers, convolutional layers allow to use advanced activation functions such as the rectified linear unit (ReLU).

○ (D) Multi-channel colour images can only be processed using convolutional layers.

○ (E) In convolutional layers the number of parameters needed by the network to process images scale much better with increasing image size.

**Solution:** (B) and (E) are correct.